MineOcclude: Separating Visual Evidence, History, and Motion Predictability

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Abstract

MineOcclude separates visual evidence, image history, and motion predictability through paired Minecraft captures with continuous position targets. Frozen MineWorld with verified RGB inputs achieves 0.083–0.107 blocks RMSE in air and glass and 1.600 under stone on 25 central trajectories. Across two layouts, 64 training and 64 test families provide varied paired motions. An exploratory follow-up reuses these features and families: direct contrast training reduces stone error at ages 1–10 from 1.176 to 0.280 for MineWorld, from 1.125 to 0.183 for history pixels, and from 1.019 to 0.179 for VideoMAE. Pixels, VideoMAE, and an image-history control outperform zero contrast (0.265 blocks); MineWorld does not clearly do so. Pixels and VideoMAE also outperform a privileged direction/age reference early, but this reference beats every image readout over all 19 ages. Source-selected scaling reduces extreme transfer errors without restoring reliable transfer. These findings characterize probe objective and scaling sensitivity in reset 16-frame windows with predictable schedules, not persistent object memory.

1 Introduction

Agents acting under partial observation must estimate state from incomplete evidence [1]. Learned world models can support this through dynamics and memory [2–6]. Interactive models generate visual continuations from images and actions [7–11], but plausible outputs do not establish which physical variables remain accessible in their features [12, 13].

Occlusion offers a direct test of the relationship between visual evidence and physical state [14–17]. Yet position may remain predictable from a fixed trajectory even when no useful object evidence is retained. Conversely, a failed probe may reflect its training objective rather than the absence of information. A probe is a simple predictor fitted to a model's features; its accuracy depends on the target, capacity, and evaluation design [18–20]. Decoding is not, by itself, evidence of calibrated uncertainty [21, 22] or causal use during generation [23–25].

MineOcclude pairs the same Minecraft sheep motion, camera, actions, and target times across visual conditions. Server state supplies continuous coordinates aligned to rendering, without estimating labels from the images being probed. The central study compares a readout fixed during development across air and glass, full occlusion, and a controlled opacity overlay. A separate motion study adds acceleration, stops, and reversals in paired trajectories across two layouts. The variants differ in their visible approaches as well as later motion, allowing an estimator to use evidence about their different schedules.

Our contributions are a paired evidence protocol with grouped evaluation and metadata references; comparisons of absolute position and paired variation using equal histories and horizons; and tests showing that probe objectives and source scaling change conclusions drawn from the same features.



Figure 1: Recorded central inputs at a common target time: air, white glass, light blue glass, brown glass, and stone. Motion and evaluation times are paired. Relative sheep image energy is $V = 1, 0.8131, 0.4260, 0.1857, 0$, respectively. The materials also change appearance, so this ordering alone is not a controlled transparency manipulation.

Direct contrast training recovers differences missed by absolute-position probes, while transfer tests expose scaling amplification. This is a controlled evaluation of existing readouts, not a new probing algorithm or a demonstrated memory mechanism.

The public code repository and OSF archive provide code, fitted probes, predictions, targets, splits, and features for the direct-contrast, image-control, and transfer-scaling analyses, with instructions for reconstructing the statistics and refitting the readouts.

2 Related Work

Latent dynamics models connect perception to planning [3, 4, 26, 27], while object-based models explicitly represent entities and interactions [28–30]. IntPhys, CLEVRER, PLATO, Physion, Physion++, and WorldBench assess persistence, interactions, or physical concepts [14, 31, 16, 32, 33, 17]. Physics-IQ and related evaluations distinguish visual quality from physical validity [12, 34, 35]. MineOcclude complements behavioral tests with aligned continuous position readouts under controlled evidence changes.

Probing studies recover environmental state, scene variables, and belief geometry from learned representations [36–40]. The closest physical-representation studies go beyond a single decoding score. Joseph et al. [41] combine layer probes, subspace geometry, local decoding, and attention interventions. Shin et al. [42] examine hidden state tracking and extrapolation in an action-conditioned visual Shell Game. Their evaluation concerns state tracking beyond visible observations; ours resets the cache for every 16-frame window and reads the current visual slice. Interventions and concept methods address causal use more directly [43, 24]. Our distinct contribution is a paired evidence protocol that tests absolute position, relative motion, and the sensitivity of fitted readouts. Frozen VideoMAE provides an additional video encoder comparison [44], not another world model family.

3 Methodology

3.1 Paired captures and continuous targets

The central cohort contains 25 seeded trajectories distinct from development, with 13 moving left to right and 12 right to left. Every trajectory is captured in air, white glass, light blue glass, brown glass, and stone (Figure 1). A trajectory is the independent identity; frames from it are repeated observations. Motion, camera, actions, and lighting are shared across its five captures. At 20 Hz, the nominal speed is 2.0 blocks/s with process noise 0.05 blocks^2/s. Each 180-frame capture contributes the same 19 target times, selected when the full sheep extent would be behind stone. No trajectory is selected by decoding performance.

The target $x(t) = \sum_{k=0}^{8} h_{d,k} x_{\text{server}}(t-k)$ is position parallel to the wall, aligned to rendered motion by a calibrated direction-specific finite impulse response filter. Its coefficients are fixed before these evaluations. Geometry determines full occlusion using the whole object extent. Relative image evidence V compares sheep-present and sheep-absent image energy with air, using a shared mask and a background correction. It is not a visibility probability or model confidence. Appendix A specifies the alignment and evidence measurement.

The paired motion study has 64 training and 64 test families, balanced across two layouts, directions, and four schedules: velocity, acceleration, stop/resume, and reversal. Each family has two variants,

Table 1: Independent units and reuse. A paired family contains two motion variants. The direct-contrast, image-control, and transfer-scaling analyses reuse the paired-motion train/test split; they do not provide independent confirmation.

Cohort	Units	Role
Development	25 previously collected identities	Layer and spatial readout selection.
Central	25 identities distinct from development	Five materials, RGB, history, opacity, transfer; five outer folds.
Supporting studies	75 + 96 separate identities	Geometry/speed changes and history substitutions; Appendix C.2.
Paired motions	64 train + 64 test families	Two layouts; absolute fits, direct contrasts, positive controls, transfer sensitivity.

each recorded in air and stone. Variants share the server position at occlusion entry but differ in visible approach and later motion; their temporally aligned targets need not coincide. Because alignment averages recent states, early contrast targets can include differences from before full occlusion. Layout 04 changes sheep depth and wall geometry relative to layout 01. Schedules are fixed once sampled, not unpredictable hidden branches. Admission checks and engineering amendments preceded test capture and fitting. The central study, paired study, and supporting cohorts described in Appendix C.2 are kept separate (Table 1).

3.2 Representations and grouped fitting

Frozen MineWorld `700M_16f` processes the 16 recorded image/action pairs at $[t-15, \dots, t]$, spanning 0.75 s [11]. The transformer cache is reset and positions start at zero for every target. Only the current visual slice is read after the full window is processed. Layer 19 and coarse 2×6 pooling were chosen during separate development and remain fixed. Pooling the 14×24 token grid yields 24,576 features. Every primary result uses verified RGB input resized to 224×384 and normalized to $[-1, 1]$. Patch tests check the actual tokenizer input; BGR-input results are reported separately in Appendix C.1.

Current pixels use one 48×28 RGB image; history pixels concatenate all 16. Further comparisons use current pixels plus consecutive image differences, optical flow [45], and frozen VideoMAE-S features [44]. Current image plus all differences is an invertible linear reparameterization of history pixels, so its comparison concerns scaling and regularization, not extra information. Recorded actions are constant no-ops. Image predictors receive neither direction, age, nor server state. Matching history does not match resolution, pretraining, or capacity.

Native ridge probes are fitted and tested within a condition [46]. Central evaluation uses five outer identity folds and four inner rotations. Paired motion evaluation keeps both variants together, with 32 training and 32 test families per layout and four folds within training. All fits and tuning use the 19 training ages; ages 1–10 are the primary test interval. Feature masks, scaling, and penalties are learned from training data only, following grouped and nested evaluation principles [47–50]. Standard ridge uses $\lambda \in \{0.1, 1, 10, 100, 1000\}$, selected by mean validation RMSE.

3.3 Training for absolute position and paired differences

For family f, define $\Delta h_{f,t} = h_{f,1,t} - h_{f,0,t}$ and $\Delta x_{f,t} = x_{f,1,t} - x_{f,0,t}$. The absolute-position approach subtracts the two variants' predicted positions. The direct-contrast approach instead fits feature and target differences:

$$\widehat{\Delta x}_{f,t} = w^\top S_{\text{train}}^{-1} \Delta h_{f,t}, \qquad w = \arg\min_{w} \sum_{f,t \in \text{train}} \left(w^\top S_{\text{train}}^{-1} \Delta h_{f,t} - \Delta x_{f,t} \right)^2 + \lambda \|w\|_2^2.$$

The feature mask and diagonal scales S_{train} come from the absolute features of training variants. There is no intercept, so swapping variants reverses the prediction. The same linear function class on feature differences is available in both analyses, but the target and penalty selection now optimize contrast error. Both input histories are required: this is a paired diagnostic, not an online estimate from one trajectory.

Predicting zero difference provides a target-scale reference. A separate cubic regression learns the expected signed contrast from direction and age on training families. It exploits the protocol's ordered pair and predictable schedules, and is explicitly privileged. It differs from subtracting a common absolute direction/age predictor, which necessarily yields zero. Last-position and constant-velocity oracles use server history before occlusion. These are predictability references, not equivalent-input baselines [51, 52].

An image-based positive control extracts the largest magenta component in each of the same 16 frames. Nine normalized position, extent, area, and missingness features per frame feed a ridge or radial basis function (RBF) estimator. Small partial detections are retained and empty windows are not excluded. Fits use training families only; the contrast RBF estimator reverses sign when the pair is reversed. No coordinates, age, direction, variant label, or paired air image enters the stone features. This deliberately benchmark-specific estimator tests recoverable motion evidence, not generic tracking (Appendix B).

3.4 Transfer and controlled opacity

A transferred probe keeps its source fit unchanged in a destination condition. We compare standardization with two source-selected alternatives: a standard-deviation floor $s'_j = \max(s_j, \tau \operatorname{median}_{s_k>0} s_k)$, with $\tau \in \{0.01, 0.1, 1\}$ jointly tuned with ridge penalty; and centered ridge without variance scaling, with penalties $10^{-6}, 10^{-4}, \ldots, 10^{6}$. Exactly constant source features remain excluded. Selection never uses destination outcomes. We report the central five-material matrix and air/stone transfers for all three paired-study representations, including native diagonals.

To separate opacity from material appearance, one fixed stone texture is blended into a geometry-defined mask in every central air image at opacity $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$. Targets, folds, and pixels outside the mask stay fixed. Native fitting and unchanged air transfer are compared. This is a controlled image manipulation, not a physical model of glass.

3.5 Metrics and analysis status

RMSE is computed over the specified ages within identity and then averaged. For absolute errors in paired motions, the two variant RMSEs are averaged within family. Contrast RMSE is computed directly from $\widehat{\Delta x} - \Delta x$ within family. These are means of per-unit errors, not a pooled frame RMSE. Two-sided 95% percentile intervals use 10,000 bootstrap draws [53]. Central draws resample whole identities; paired draws resample whole families within layout and motion type. Error differences use the same sampled units.

Direct contrast fits, image controls, and transfer scaling are exploratory analyses motivated by earlier results on the same test families. Settings preceded their own outcomes; test families never enter fitting or hyperparameter selection. This prevents fitting leakage, not cohort reuse. Intervals condition on fitted probes, omit refitting uncertainty, and lack multiple-comparison adjustment. Ages 1–19 and individual layouts supplement, rather than replace, the primary interval.

4 Experiments

4.1 Visible decoding and history on the central cohort

The fixed MineWorld readout achieves 0.083–0.107 blocks RMSE in air and glass, and 1.600 under stone (Table 2). Its stone interval is $[1.496, 1.702]$. Giving pixel ridge the same 16 images lowers error from 3.424 to 1.818; MineWorld retains an advantage of 0.218 blocks, interval $[0.173, 0.265]$. Frame differences, optical flow, and VideoMAE reach 1.781, 1.757, and 1.767. Current pixels are more accurate than MineWorld in every visible material, so the central advantage is specific to the tested hidden-history comparison.

Metadata remains stronger under stone: direction and age give 0.199 blocks, oracle velocity 0.403, and oracle last position 0.914. Accurate position is therefore not sufficient evidence of object memory. MineWorld stone errors across ages 1–5, 6–10, 11–15, and 16–19 are 0.428, 0.369, 1.938, and 2.603. At age 16 the last observation before occlusion has left the reset window. Earlier deterioration is compatible with weak localization evidence, while later error cannot diagnose memory outside

Table 2: Central cohort, all 19 ages: mean identity RMSE in blocks. All predictors share target rows and grouped folds; image resolution and estimator capacity differ.

Predictor	Air	White glass	Blue glass	Brown glass	Stone
MineWorld	0.085	0.083	0.094	0.107	1.600
Current pixels	0.050	0.049	0.054	0.054	3.424
History pixels	0.088	0.180	0.066	0.060	1.818
Frame differences	0.433	0.183	0.141	0.142	1.781
Optical flow	0.143	0.278	0.281	0.294	1.757
VideoMAE-S	0.068	0.082	0.080	0.078	1.767

Table 3: Stone absolute-position errors at common horizons. Δ is MineWorld minus history pixels with paired 95% interval. Central summaries use 25 identities; paired summaries use 64 families pooled or 32 per layout.

Cohort / layout	Ages	MineWorld	History pixels	VideoMAE	Δ [95% CI]
Central	1–10	0.418	0.620	0.529	-0.201 [-0.376,-0.039]
Central	1–19	1.600	1.818	1.767	-0.218 [-0.265,-0.173]
Paired / pooled	1–10	0.768	0.635	0.656	0.133 [0.066,0.205]
Paired / layout 01	1–10	0.592	0.731	0.632	-0.139 [-0.210,-0.065]
Paired / layout 04	1–10	0.945	0.539	0.680	0.406 [0.287,0.529]
Paired / pooled	1–19	1.685	1.718	1.712	-0.033 [-0.058,-0.007]

the supplied window. Separate native fits with one, eight, and sixteen recorded pairs give 3.424, 2.726, and 1.600 blocks; longer history helps on these identities, without isolating a persistent-state mechanism.

4.2 Varied motion and absolute position accuracy

On 64 test families at ages 1–10, stone error is 0.768 for MineWorld, 0.635 for history pixels, and 0.656 for VideoMAE. The paired excess of MineWorld is 0.133 blocks over pixels, interval $[0.066, 0.205]$, and 0.112 over VideoMAE, $[0.037, 0.187]$. This pooled ordering is driven by layout 04: MineWorld is more accurate than pixels in layout 01 and less accurate in layout 04 (Table 3). The VideoMAE difference in layout 01 is inconclusive.

Layout 04 keeps the camera fixed but changes sheep depth and wall placement, altering apparent object scale and occlusion geometry. The reversal may therefore reflect sensitivity of the encoder, spatial pooling, or fitted probe to image geometry, rather than motion information alone. Work on CNN shift invariance motivates this concern [54, 55], but does not establish its cause in MineWorld. The two layouts change several factors together and are not a controlled translation or camera-shift test.

Across all 19 ages, MineWorld has the lowest pooled mean of the three (1.685 versus 1.718 and 1.712), but its VideoMAE difference includes zero. This secondary result does not replace the primary interval. Direction and age still predict early stone position at 0.211 blocks; adding motion type reaches 0.154 and oracle velocity 0.328. The varied schedules neither remove predictability nor isolate rendering changes. Comparisons across cohorts therefore do not isolate motion diversity as the cause of changed model rankings.

4.3 Training for differences changes the paired result

The absolute-position fits give poor paired differences: 1.176 blocks for MineWorld, 1.125 for history pixels, and 1.019 for VideoMAE, compared with 0.265 for zero difference. Fitting directly to contrasts substantially changes the result on the same test families (Table 4). MineWorld reaches 0.280, pixels 0.183, and VideoMAE 0.179. Each improves on its corresponding absolute-fit difference with a paired interval wholly below zero. Pixels and VideoMAE now also improve on zero difference in both layouts. MineWorld's excess over zero is 0.015 blocks with interval $[-0.018, 0.050]$, which establishes neither an advantage nor equivalence to the reference.

Table 4: Stone paired differences, ages 1–10, averaged over 64 test families. "Absolute fit" subtracts separately predicted positions; "Direct fit" trains on feature and target differences. The last column compares direct fit with zero difference (0.265). The image controls use detected image features, not server state.

Predictor	Absolute fit	Direct fit	Excess over zero [95% CI]
MineWorld	1.176	0.280	0.015 [-0.018, 0.050]
History pixels	1.125	0.183	−0.082 [-0.097, -0.067]
VideoMAE	1.019	0.179	−0.086 [-0.100, -0.071]
Image control (ridge)	0.830	0.210	−0.055 [-0.065, -0.044]
Image control (RBF)	0.826	0.206	−0.059 [-0.070, -0.049]

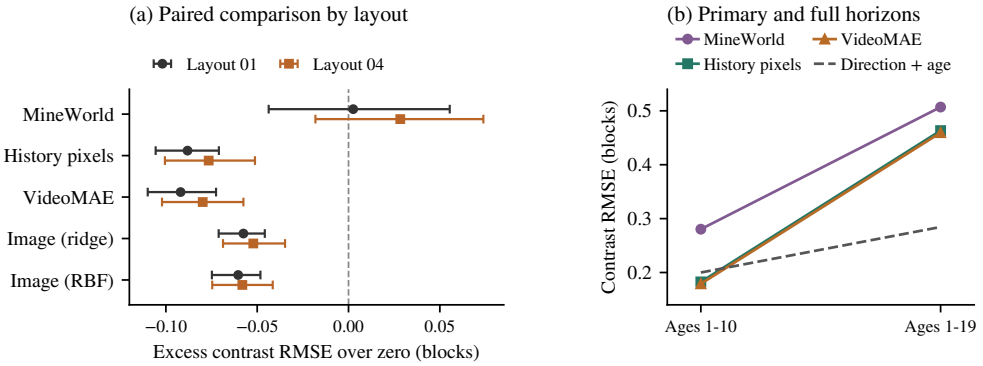


Figure 2: Direct contrast fitting under stone. (a) Excess error over zero difference at ages 1–10, separately for 32 test families in each layout; bars are paired 95% intervals. Negative values favor the readout. These early input windows can retain observations from before full occlusion. (b) Contrast RMSE at the primary and full horizons. The privileged direction/age reference becomes stronger than every image predictor over all 19 ages. These aggregates do not establish accurate tracking at each late age.

The image control provides a positive result beyond counting visible pixels. Its ridge and RBF contrast errors are 0.210 and 0.206, respectively, with improvements over zero in each layout. Current-air localization is 0.075 blocks after an affine calibration fitted only on training air targets. At ages 1–10, 97.5% of stone windows contain an accepted detection; the mean over all windows is 7.30 detected frames per window. All windows remain in evaluation, including the 2.5% with none. Separately fitted absolute image controls reach 0.444 and 0.438 blocks, compared with MineWorld's 0.768 and metadata's 0.211. This supplies a usable history comparator while leaving a gap to privileged trajectory prediction. It is not a generic tracker or a claim that every window is informative.

A cubic direction/age predictor trained for the ordered contrast reaches 0.200 blocks. Pixels and VideoMAE improve on it by 0.017 and 0.021, with intervals $[0.003, 0.031]$ and $[0.007, 0.036]$. The image controls do not clearly improve on this stronger reference. Successful contrast decoding establishes recoverable variation in the supplied windows, not persistent state tracking: the variants have different visible approaches, schedules remain predictable, and early aligned targets can contain motion before full occlusion. The key finding is that a failed absolute-position objective did not establish an absence of linearly decodable paired information in those same features. This follow-up reuses the test families that motivated it, so the result is exploratory rather than independent confirmation.

At ages 1–19, pixels and VideoMAE still improve on zero (0.497), while MineWorld remains inconclusive. However, the direction/age contrast reference reaches 0.284, better than every direct fit (0.459–0.507; Figure 2). Thus the advantage over this metadata reference is confined to the early interval. No accepted stone detections remain at ages 16–19; an aggregate improvement over zero does not establish tracking after the informative observations leave the window.

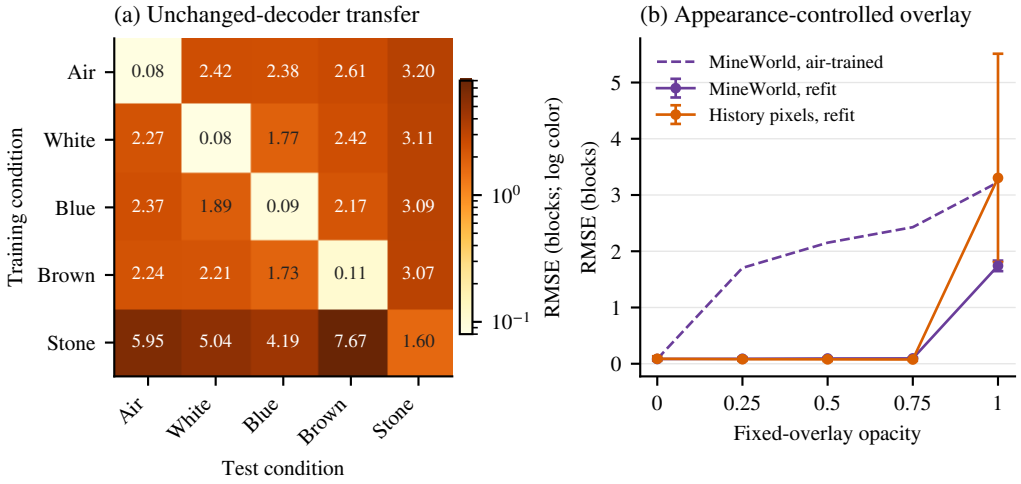


Figure 3: Central transfer with standard feature scaling and controlled opacity. (a) Source probes and their fitted preprocessing are applied unchanged to test identities in each destination; color is logarithmic. (b) A single texture and mask at varying opacity. Native fits use training folds at each level; the dashed curve keeps the air fit fixed. Intervals resample identities. Alternative-scaling results appear in Table 5.

4.4 Transfer depends on source scaling

In the central matrix, the air-trained probe with standard feature scaling gives 2.424, 2.384, 2.609, and 3.203 blocks on the glass conditions and stone (Figure 3). Every off-diagonal is worse than the destination's native fit. The source-selected alternatives do not restore this air-to-stone mapping: the floor and unscaled fits give 3.228 and 3.277. Thus accurate native decoding does not imply a stable fitted mapping across appearance changes.

The fixed overlay gives native errors of 0.085, 0.088, 0.093, 0.095, and 1.742 as opacity increases. Unchanged air transfer reaches 1.704 already at 25% opacity and 3.226 at full opacity. Partial evidence remains decodable after refitting, although the unchanged air-trained mapping is not stable. Zero opacity exactly reproduces air pixels and features; full opacity masks all detected sheep pixels in scored frames. This controls the overlay appearance, not all physical effects of glass.

The paired motion study reveals substantial estimator sensitivity (Table 5). Stone-to-air history-pixel RMSE falls from 233.397 to 39.282 with a variance floor and 34.413 without variance scaling. VideoMAE falls from 555.042 to 333.584 under the latter fit. Saved predictions and feature statistics for the standard-scaling fits were independently replayed: in layout 04, the 99th percentile of absolute standardized values rises from stone to air test inputs, both scaled with stone training statistics, from 6.20 to 1064.76 for pixels and 3.87 to 564.78 for VideoMAE. Small source variances amplify changed destination inputs. Some extreme failures therefore depend strongly on scaling, but none of these tested alternatives generally restores accurate transfer. MineWorld's stone-to-air errors do not improve in this paired cohort. All methods were selected using source data only; none was chosen by its destination score.

Native fits also change with the estimator. MineWorld stone error falls from 0.768 to 0.685 without variance scaling; history pixels reach 0.588. The strong pixel stone-to-air improvement therefore does not result from a universally more accurate decoder: its transferred 34.413 blocks remain far above native air error of 0.055. In the central matrix, all 20 off-diagonal errors exceed the corresponding destination-native error under each of the three fitting schemes. These limited source-selected alternatives strengthen the finding that the tested mappings are unstable, without ruling out a different decoder, training distribution, or representation.

Table 5: Source-selected sensitivity on paired motion families, ages 1–10, pooled over layouts. Mean family absolute-position RMSE in blocks. "Standard" uses source training means and standard deviations. "Floor" tunes the source variance floor and ridge penalty; "Unscaled" centers features without variance normalization. Native rows show that poor transfer coexists with usable source fits. No alternative is selected by destination performance.

Source → destination	Predictor	Standard	Floor	Unscaled
Air → air (native)	MineWorld	0.080	0.078	0.077
	History pixels	0.061	0.052	0.055
	VideoMAE	0.063	0.063	0.063
Air → stone	MineWorld	3.603	3.624	3.637
	History pixels	11.516	3.186	3.387
	VideoMAE	3.473	3.473	4.610
Stone → stone (native)	MineWorld	0.768	0.723	0.685
	History pixels	0.635	0.597	0.588
	VideoMAE	0.656	0.656	0.682
Stone → air	MineWorld	5.597	5.952	7.954
	History pixels	233.397	39.282	34.413
	VideoMAE	555.042	555.042	333.584

4.5 Interpretation, reproducibility, and scope

Absolute accuracy, paired variation, and transfer test different properties. Failed absolute-fit contrasts do not exclude linear contrast decoding; successful controls establish neither memory nor planning benefit. Predictions use reset 16-frame windows, two per contrast, rather than continuous state retention.

Scope is limited to one sheep, two layouts, several schedules, and one world model family: MineWorld 700M, with 300M only in a supporting diagnostic. VideoMAE is an encoder comparison. Other entities, environments, and model families remain untested; resolution, pretraining, and capacity are unmatched. Layout changes do not isolate spatial sensitivity, rendering differences confound cohort comparisons, and the overlay is synthetic. Exploratory analyses reuse previously examined test families; grouped separation prevents fitting leakage but does not make their conditional bootstrap intervals confirmatory.

Future work should compare continuous and reset inference on new families with observable cues or actions determining hidden motion. Random hidden branches with identical images and actions instead test ambiguity: their realization cannot be inferred without additional evidence.

The release includes features, targets, folds, fits, predictions, code, and model provenance. Checks covered the encoder interface on 512 captures, limited fresh neural extraction, exact statistical reconstruction, replay of 108,870 predictions within 10^{-11} blocks, and the transfer diagnostic. Downloads without owner credentials passed hash checks. These validate artifacts, not independent replication; Appendix D provides instructions and separately acquired assets.

5 Conclusion

MineOcclude separates image evidence, motion predictability, and the behavior of fitted position readouts. On paired motions, direct contrast training recovers variation with history pixels, VideoMAE, and an image-based control that absolute-position fits miss. MineWorld improves substantially but does not clearly outperform zero contrast. Source-selected scaling reduces some severe transfer errors without recovering a stable decoder. These findings support contrast objectives, equal-history controls, and preprocessing sensitivity checks for physical state evaluation. They describe tested reset-window readouts, not persistent object memory or a general property of world models.

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A Measurement and cohort specifications

A.1 Geometry, motion, and alignment

Minecraft 1.16.5 records 640×360 images. In the central setting, the sheep follows $x \in [-8, 8]$ at $(y, z) = (4, 6)$; the camera eye is $(0.5, 5.6200000048, -5.5)$ with zero yaw/pitch/roll and 70° field of view. The wall spans $[-3, 4] \times [4, 7] \times [2, 3]$. At $\Delta t = 0.05$ s,

$$x_t = \mathrm{clip}_{[-8,8]}\left(x_{t-1} + d\,2.0\,\Delta t + \sqrt{0.05\Delta t}\,\epsilon_t\right), \qquad \epsilon_t \sim \mathcal{N}(0, 1).$$

Drift is accumulated before noise in binary64. NumPy is seeded by identity; starts use a separate seed obtained by XOR with `0xE3E3E3E3`. The roster records exact paths and target times. Actions are constant: buttons false, camera increments zero, and unchanged hotbar.

The alignment taps below are in increasing lag order and are normalized by their sum at loading. Target reconstruction using `math.fsum` agrees with the separately checked reconstruction helper within 2×10^{-15} blocks. Calibration predates the reported probe outcomes.

Lag	Left to right	Right to left
0	0.0552210582	0.0155128136
1	0.0955547802	0.1420986717
2	0.1729033607	0.3819949723
3	0.2562377030	0.2529919934
4	0.1725945773	0.1314006058
5	0.1502955787	0.0092585384
6	0.0283805770	0.0336603539
7	0.0327627758	0.0330820509
8	0.0360495890	0.0000000000

If t_0 is the last frame before full occlusion, age a uses $[t_0 + a - 15, \ldots, t_0 + a]$. Age 15 includes t_0; age 16 does not. The cache resets for every target. Geometry identifies possible evidence; detected image content and estimator performance are separate checks of what is actually usable.

A.2 Evidence and opacity

At nine placements $x = -2, -1.5, \ldots, 2$, sheep-present/absent capture blocks use orders P–A–A–P and A–P–P–A, with at least eight settled images per state and block. RGB values $s = v/255$ are linearized as $s/12.92$ below 0.04045 and $((s + 0.055)/1.055)^{2.4}$ otherwise. Pixelwise medians give present and absent images P, A. For mask M, $D = \sum_{u \in M} \|P(u) - A(u)\|_2^2$. Background energy N compares the median first and last halves of absent images, omitting an odd middle frame. Excess energy is $E = \max(D - N, 0)$.

The common mask unions air residuals above the maximum absent-half residual across blocks and dilates by a two-pixel square neighborhood. Ratios E_c/E_{air} are computed within block and placement, then median aggregated over blocks and placements. Their median is V_c; $1 - V_c$ is not a calibrated physical transparency. Stone has zero positive residual under this measurement, not proof that every indirect cue is absent.

The opacity control takes its fixed stone texture from frame zero of the first predetermined central stone capture. The mask is the convex hull of the eight projected wall corners. Within it, $I_\alpha(u) = \mathrm{round}((1 - \alpha)I(u) + \alpha W(u))$; outside it, pixels are unchanged. All 16 images are transformed. The mask covers detected sheep pixels in all 475 scored air frames. Zero opacity is bit-identical to air pixels and MineWorld features. Native and all transferred opacity predictions are retained; the main figure shows native fits and air transfer without choosing a favorable subset.

A.3 Paired motion schedules

Each partition has 32 families per layout and eight per motion type, balanced by direction. Two variants and two materials give 512 captures and 9,728 scored rows overall. Layout 04 changes sheep depth to $z = 8$ and wall depth to $[3, 4]$, with the same camera. Both apparent scale and occluding geometry change.

At 20 Hz, the low speed parameter is uniform on $[1.4, 1.9]$ blocks/s, except reversal uses $[1.6, 1.9]$. The high parameter is uniform on $[2.8, 3.5]$, except constant velocity uses $[2.65, 2.8]$. Acceleration magnitude is uniform on $[0.8, 1.5]$ blocks/s^2, negative for the low variant and positive for the high, clipping speed to $[0.25, 4.5]$. Stop/resume uses eight moving and four stopped ticks per 12-tick cycle, phases zero and six, and moving speed 1.5 times the sampled parameter. Reversal adds a sinusoid of amplitude 3.8 blocks/s and period 12 ticks with phases zero and six. Motion stops during ticks 0–19 and after tick 109. Starts align the first occlusion position within direction and layout. The roster contains exact schedules, targets, seeds, and folds.

Admission checks schedule replay, state/image joins, completed image writers, wall construction, visible air sheep, and no detected sheep pixels in scored stone frames. Air-image checks require centered position discrepancy below 0.15 blocks and median offset below 0.30. Entry requires at least three prior images above the 20-pixel admission threshold. Pilots cover all layouts, directions, variants, and motion types. Before test capture or fitting, engineering amendments bounded high velocity, extended some approaches, and raised the low reversal bound to satisfy unchanged checks. Captures superseded by these pre-test engineering amendments are excluded from the final roster; exclusions were not based on test outcomes.

B Features and readout fitting

B.1 Image features and paired objectives

A MineWorld image/action pair has 336 visual and 11 action tokens. Sixteen pairs contain 5,552 tokens of width 2,048. The current visual slice is $[5205{:}5541)$; action tokens are excluded from pooling. The 2×6 averages yield 24,576 features. Tokenizer caching retains no temporal transformer state. Patch tests, fresh-cache determinism, and cached/uncached equality were checked over 30 central windows spanning directions, materials, and ages.

Pixels use area resizing to 48×28. Flow uses 15 dense Farnebäck fields on 96×56 grayscale images (pyramid scale 0.5, three levels, window 15, three iterations, polynomial size 5, sigma 1.2). Fields are area resized to 24×14 and joined with current RGB pixels, yielding 14,112 features. VideoMAE-S uses its official Kinetics-400 self-supervised 1,600-epoch checkpoint [44]: 12 blocks, width 384, six heads, tubelets of two frames. RGB is resized bicubically to 224×224 and normalized with ImageNet statistics. Unmasked 14×14 tokens are averaged over eight temporal slices, then pooled to 2×6 cells, yielding 4,608 features. Neither encoder is fine-tuned.

Direct contrasts use $\Delta h = h_1 - h_0$ and $\Delta x = x_1 - x_0$, ordered by the fixed roster. Within each training fold, the mask retains columns with positive range across absolute features; standard deviations are also computed on those absolute features. Subtraction cancels the training mean. Ridge has no intercept and is selected by mean family contrast RMSE over all 19 validation ages. The full training partition is then refitted and evaluated without changing settings. This reduces the number of fitting rows relative to absolute regression because each pair supplies one difference. The comparison changes the supervised task and its validation criterion, not the stored representation.

The privileged contrast reference fits ordinary least squares on $[1, d, a, da, a^2, da^2, a^3, da^3]$, with a normalized by 19. It learns the typical ordered contrast from training families. It receives neither family-specific images nor variant motion parameters, but uses direction, age, and the known pair order. It must not be conflated with subtracting the same absolute metadata predictor on both variants.

B.2 Image-based positive control

For RGB channel values in $[0, 255]$, the fixed mask is $R > 45$, $B > 30$, $R > 1.4G$, and $B > 1.25G$. The largest 8-connected component is accepted if it has at least three pixels. Its horizontal/vertical centroid, left/right/top/bottom bounds, width, area, and observation indicator form nine features, with coordinates normalized by image dimensions and area by pixel count. Missing detections produce zeros including the indicator. Concatenating 16 frames gives 144 features.

Ridge uses the same penalty grid as neural probes. RBF uses $k(z, z') = \exp(-\gamma \|z - z'\|^2)$, with $\gamma \in \{0.1, 1, 10\}/p$, where p is the retained feature count, and penalties $\{0.01, 0.1, 1, 10\}$. Contrast fitting uses the odd kernel $k_{\mathrm{odd}}(z, z') = \frac{1}{2}[k(z, z') - k(z, -z')]$, which makes the predicted sign reverse when the pair reverses. Masks, scaling, and both hyperparameters use training families

only. Absolute RBF fitting centers features and targets; contrast fitting fixes centering offsets to zero, without assuming zero empirical means. The current-air localization check fits a separate affine map from detected horizontal centroid to aligned coordinate using training air only. It is a detector check, not input to the stone estimators.

Table 6: Image-control detector coverage on paired-study test stone windows. A detection needs three pixels. These counts are not statistical sample sizes or guarantees of precise localization. The tracker in Appendix C.2 used a different cohort and detection/extrapolation rule; its 59.4% fallback rate is not directly comparable.

Ages	Windows	No detections (%)	Mean detected frames
1–10	1280	2.5	7.30
11–15	640	59.2	0.85
16–19	512	100.0	0.00
1–19	2432	38.0	4.06

B.3 Transfer alternatives and uncertainty

The standard-scaling pipeline removes exactly constant source columns and scales each retained feature as $(h_j - \mu_j)/s_j$. The floor uses $\max(s_j, \tau \operatorname{median}(s))$ for retained features; raw ridge subtracts μ_j but divides by one. The mask policy is shared. The source selects the floor/penalty pair on source validation error, not robustness on destination observations. A small selected floor may therefore reproduce the standard-scaling result. Central fits preserve the five outer identity folds and four inner rotations; paired fits retain four training-family folds. Final source fits are applied to both air and stone, or to all five central materials. All alternatives, including native diagonals and unsuccessful transfers, are saved.

The retrospective readout analyses use 10,000 bootstrap draws with seed 94212. The absolute-fit central summaries use 20260911; transfer/comparator summaries use 94111, 94112, and 94114; horizon checks use 94115. The absolute-fit paired-motion summaries use 94113. These specify separate descriptive calculations, not independent replications. Every paired contrast is resampled by family within layout and motion type. Central folds overlap across fits; intervals in all studies condition on the fitted models. No bootstrap draw refits a probe or repeats model selection.

Table 7: Direct stone contrast fits by layout. Each cell is RMSE in blocks. The full-horizon result is secondary. Layout-specific primary intervals are in Figure 2; pooled error intervals and every air comparison are in the analysis files.

	Ages 1–10		Ages 1–19	
Predictor	Layout 01	Layout 04	Layout 01	Layout 04
MineWorld	0.266	0.295	0.506	0.508
History pixels	0.175	0.190	0.465	0.462
VideoMAE	0.171	0.187	0.458	0.461
Image control (ridge)	0.206	0.215	0.474	0.477
Image control (RBF)	0.203	0.209	0.474	0.477
Zero difference	0.263	0.267	0.499	0.495
Direction + age	0.198	0.202	0.284	0.285

C Development and supporting studies

C.1 Development and RGB verification

Development sampled layers $\{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 19\}$ with global and coarse 2×6 pooling. Layer 19 advanced to global, coarse, and Best-K refinement with $K \in \{1, 4, 16, 64\}$ selected on training identities. Its coarse readout was retained for RGB evaluation without reselection. These 25 development identities used BGR channel order. Paired no-wall and hidden populations each had 475 rows; a separate wall-visible control had 846 and a different target range. Visible controls established usable decoding, not a matched effect-size comparison with hidden rows.

The nine development-stage evaluation criteria concerned provenance/leakage; mean improvement and its lower bound; macro R^2 and its lower bound; RMSE ≤ 0.50; fold stability; both directions; phase stability; visible controls; and selection-aware $p_A \leq 0.05$. Only the absolute R^2 and RMSE criteria failed (macro $R^2 = -13.8263$, RMSE 2.0711). The 49-draw refinement null [56] had add-one minimum resolution 0.02 [57]; the record does not establish repetition of the entire 22-cell screen. These development-stage criteria are not the inference rules for the central, paired-motion, or supporting studies reported here.

Predictions generated with BGR inputs are labeled separately from RGB predictions. Replaying BGR on the same runtime as RGB gives stone RMSE 1.693 versus 1.600, a reduction of 0.093 with interval $[0.040, 0.153]$. The separately recorded BGR execution gave 1.645 on its own runtime; it is distinct from the matched-runtime replay. The conditions have separate prediction records. Primary statements in this manuscript use the verified RGB execution. The central stone trajectory $R^2 = -9.324$ uses each test trajectory's own mean in its denominator; improvement over a constant learned on training identities uses another reference. Beating that training constant does not imply resolving the smaller variation within a test path.

C.2 Geometry/speed study and history interventions

A separate geometry/speed study captured 25 identities per setting: the central configuration, layout 04, and higher speed (3.0 rather than 2.0 blocks/s), with five grouped folds within setting. Starts have magnitudes 7.3–7.9; admission requires 19 consecutive target times, complete histories, no target clamp, and target standard deviation above 0.10. Its results remain separate from paired families.

The history diagnostic used 96 separate identities, 24 training and 24 test per layout, with speeds from $[0.8, 1.6]$ or $[2.4, 3.2]$, balanced by direction. A predetermined reciprocal slow/fast donor shares layout and direction. Swapping replaces the preceding 15 images but retains the recipient's current image; removing history repeats that current image. Fits use correct training histories. Donor-pair resampling is used for intervention differences, and identity resampling for ordinary comparisons (seed 930119).

At ages 1–10, stone RMSE was 1.603 for 700M ridge, 1.588 for 300M ridge, and 1.915 for history pixels; the 700M-minus-pixel interval $[-0.784, 0.013]$ includes zero. A nonlinear probe used training PCA to 64 components, 32 tanh units, penalties $\{0.001, 0.1\}$, and three fixed seeds. Its initial-budget fits had convergence warnings. A separately fixed budget check extended L-BFGS to 10,000 iterations and 250,000 function evaluations without reopening selection; all final fits converged and gave 1.240/1.341 for 700M/300M. Both budgets remain recorded. This changes size within one family, not world model diversity.

For 700M, removing history increased error by 2.314 blocks, interval $[2.068, 2.560]$; swapping changed it by -0.094, $[-0.144, -0.047]$. Prediction shifts agreed with donor-minus-recipient position on 32.1% of rows, indicating sensitivity rather than reliable donor tracking. The image tracker evaluated on this history-diagnostic cohort had error 2.918 and used a training-mean fallback on 59.4% of rows. Its stricter detection/extrapolation rule and distinct cohort prevent a direct coverage comparison with the paired-motion image control in Appendix B.2. Geometry transfer for 700M gave air/stone errors 1.934/2.675; affine calibration using six visible destination training identities gave 0.271/3.177, compared with 0.111/1.653 for native fitting. Complete predictions for these supporting studies are retained in a separate artifact snapshot rather than reproduced as appendix tables.

D Reproduction, access, and execution

The input collection for the central and paired-motion readout analyses covers 512 complete motion capture units, central image-window unions, the three feature sets used here, targets, folds, absolute-position comparisons, fitted probes and predictions, source code, and dependency specifications. Its ten prepared data archives total 8.4 GB compressed and share a SHA-256 manifest. Pretrained weights, environments, and a Minecraft installation are acquired separately. This collection supports these readout analyses, but does not include every development or supporting cohort; its full capture export is distinct from the published statistical package described below.

Portable commands verify the manifest, reconstruct statistics, replay predictions, recompute source feature diagnostics, refit contrasts and transfer alternatives, and reextract image detections. An

encoder callable receives only chronological RGB frames and an allowlisted action dictionary, without identifiers, absolute frame indices, targets, direction, or schedule metadata. A spatial RGB example was run over all 512 units through grouped fitting. It is an integration test, not a selected scientific baseline. Encoder adapters checked freshly extracted features from 19 air and 19 stone windows for each of MineWorld and VideoMAE against saved features; these spot checks do not revalidate every saved feature extraction.

Validation reproduced the statistics exactly and replayed 108,870 predictions with maximum error 9.8×10^{-12} blocks. Refit checks of the standard-scaling absolute-position pipeline matched selected penalties and predictions within 6.8×10^{-10}. Source feature quantiles were recomputed from packaged fits and features. These independently implemented arithmetic checks establish artifact consistency, not independent human replication or a new test cohort.

The public code repository and OSF archive provide access to the code and compact statistical package for the analyses reported here. Running `python reproduce_submission.py` verifies its hashes and reconstructs the reported statistics on CPU. The repository and OSF feature and analysis archives were downloaded without owner credentials and verified against their hashes. They support statistical reconstruction and readout refitting. The data README gives prediction replay, refitting, extraction, and encoder-interface commands; `UPSTREAM_ASSETS.md` supplies pinned checkpoint sources, hashes, paths, and licenses. Statistics and saved-prediction replay require neither pretrained weights nor a GPU.

Runs use one RTX 5090 (32 GB), Ryzen 9 9950X3D, and 64 GB RAM, with WSL Python 3.10.21, PyTorch 2.7.1/CUDA 12.8, NumPy 1.26.4, SciPy 1.15.2, and scikit-learn 1.6.1. Fitting and image-processing jobs for the retrospective readout analyses cap numerical libraries at four threads and avoid concurrent large fits. Computation, assembly, and initial verification took 48.8 minutes; the final whole-bundle checksum check finished about 55 minutes after the start. No new capture campaign or full neural extraction was required. Other recorded component times include 504.8 s for central RGB extraction after loading, 57.47 min for the geometry/speed capture study, and 76.09 min for the history-diagnostic capture. Receipts document components rather than total project development cost.

MineWorld and Fabric are credited under MIT and Apache-2.0. VideoMAE is used under upstream CC-BY-NC-4.0 terms for noncommercial research. The release preserves upstream licenses and notices and redistributes neither Minecraft binaries nor pretrained weights. Account and local path labels were sanitized without changing numerical observations. Large language models were used for manuscript refining, figure generation, and as code assistants. Scripts checked the claims against the saved result files and file integrity records used for the final analysis. No LLM is an evaluated model component or source of data or labels, and saved numerical outputs and validation scripts were used to check the reported results. The authors remain responsible for the core manuscript writing, scientific claims, analysis, accuracy, and integrity of the submission.

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